

Table 1: rev_data_labels

	Bounty list	Initial list
Source and period	bounty reports accepted, flagged 2024-05-18 to 2024-05-30	LayerZero, Nansen, Chaos Labs; published 18 May 2024
Addresses in list (all chains)	151,784	803,093
Ethereum interactors in list	18,211 (4.19%)	34,545 (7.95%)
IxL	15,921	29,646
IxE	698	2,669
IxI	1,566	2,206
No provider	26	24
Overlap with the other list	0	0
Bounty reports / reporters	311 / 145	–
Label in the standard setup	Sybil (positive)	non-Sybil (negative)

Table 2: rev_ensemble_appendix

	F1	AP	AUROC
LightGBM mean	0.713 ± 0.006	0.761 ± 0.006	0.968 ± 0.002
Weighted average mean	0.713 ± 0.008	0.761 ± 0.007	0.968 ± 0.002
Average – LightGBM	-0.0003 ± 0.0053	$+0.0003 \pm 0.0020$	-0.0002 ± 0.0005
corrected p (Holm)	0.941	0.860	1.000
splits with average higher	6/10	8/10	4/10
XGBoost – average	-0.0062 ± 0.0038	-0.0081 ± 0.0028	-0.0039 ± 0.0014
corrected p (Holm)	0.320	0.028	0.030

Table 3: rev_entity_level

Entity	Entities	Sybils	Any hit	Majority	Full	Recall	Unflagged ZRO	$\geq 50\%$ unflagged
E1 gas provision tree	4,907	5,463	0.755	0.755	0.753	0.711	0.380	0.246
E2 bounty report	239	5,463	0.695	0.540	0.155	0.711	0.380	0.506
E2 bounty report (all Sybils in test)	18	20	0.056	0.056	0.056	0.050	0.982	0.944
E3 reporter	132	5,463	0.864	0.674	0.114	0.711	0.380	0.394
<i>E1 gas provision tree, by test Sybils per tree</i>								
1	4,811	4,811	–	–	–	0.763	–	–
2–5	69	193	–	–	–	0.373	–	–
6–20	22	239	–	–	–	0.075	–	–
21–100	5	220	–	–	–	0.564	–	–

Table 4: rev_family_ablation

Configuration	Features	Test F1	Δ F1	Test AP	Test AUROC	F1 lower
All features	62	0.713 ± 0.006	–	0.761 ± 0.006	0.968 ± 0.002	–
Without LayerZero transactions	40	0.647 ± 0.005	-0.066 ± 0.005	0.672 ± 0.008	0.937 ± 0.004	10/10
Without Ethereum transactions	53	0.701 ± 0.007	-0.012 ± 0.006	0.742 ± 0.006	0.963 ± 0.002	10/10
Without Gas provider	51	0.708 ± 0.007	-0.005 ± 0.006	0.758 ± 0.006	0.967 ± 0.001	9/10
Without Gas provision tree	45	0.712 ± 0.006	-0.001 ± 0.005	0.762 ± 0.007	0.968 ± 0.002	7/10
Without Provision chain	59	0.711 ± 0.007	-0.002 ± 0.002	0.761 ± 0.006	0.968 ± 0.002	8/10
Without all provision-network families	31	0.705 ± 0.005	-0.008 ± 0.008	0.758 ± 0.006	0.968 ± 0.001	7/10

Table 5: rev_label_robustness

<i>LayerZero initial list among the interactors</i>			
Initial-list addresses			803,093
Overlap with the bounty-accepted list			0
Interactors on the list (all labeled non-Sybil)			34,545
Share of the non-Sybil class			8.3%
<i>Test false positives at the validation threshold (0.600)</i>			
False positives			1,461
of which on the initial list			556 (38.1%)
Precision / if the list counted as Sybil			0.727 / 0.831
<i>Model, test set</i>	<i>F1</i>	<i>AP</i>	<i>AUROC</i>
Baseline, full test	0.719	0.770	0.968
Baseline, without initial-list rows	0.758	0.806	0.972
Clean negatives, full test	0.649	0.457	0.959
Clean negatives, without initial-list rows	0.787	0.831	0.974
Initial list relabeled Sybil (exploratory; prevalence 12.1%)	0.741	0.797	0.922

Table 6: rev_levels

Level	Unit kept whole	Test Sybils	Sharing a tree	Sharing a report
1. Address split	address	5,463	599	5,434
2. Tree split	gas provision tree	5,463	0	5,406 (10 splits: $5,402.5 \pm 22.3$)
3. Report-mates held out	tree and bounty report	899-901 per fold	0	0

Table 7: rev_mimicry

Attack, features	$k = 1$	$k = 3$	$k = 5$	$k = 10$
No attack	0.711	–	–	–
M1 marginal, all features	0.410 ± 0.006	0.074 ± 0.002	0.012 ± 0.001	0.000 ± 0.000
M2 profile copy, all features	0.410 ± 0.006	0.114 ± 0.001	0.036 ± 0.003	0.005 ± 0.001
M1 marginal, cheap only	0.413 ± 0.003	0.113 ± 0.002	0.025 ± 0.002	0.003 ± 0.001
M2 profile copy, cheap only	0.413 ± 0.003	0.107 ± 0.003	0.043 ± 0.002	0.011 ± 0.001
M1 marginal, cheap + structural	0.413 ± 0.003	0.113 ± 0.002	0.025 ± 0.002	0.003 ± 0.001
M2 profile copy, cheap + structural	0.413 ± 0.003	0.107 ± 0.003	0.043 ± 0.002	0.011 ± 0.001

Table 8: rev_mimicry_controls

Attack scope	$k = 1$	$k = 3$	$k = 5$	$k = 7$	$k = 10$
No attack					0.711
<i>M1 marginal</i>					
All features (SHAP order)	0.410 ± 0.006	0.074 ± 0.002	0.012 ± 0.001	–	0.000 ± 0.000
Cheap only (SHAP order)	0.413 ± 0.003	0.113 ± 0.002	0.025 ± 0.002	–	0.003 ± 0.001
Strict cheap (SHAP order)	0.571 ± 0.003	0.254 ± 0.004	0.160 ± 0.004	0.137 ± 0.007	–
Structural only (SHAP order)	0.690 ± 0.000	0.608 ± 0.003	0.524 ± 0.001	–	0.483 ± 0.003
Random k of all 62	0.674 ± 0.075	0.570 ± 0.165	0.419 ± 0.177	–	0.235 ± 0.202
Random k of cheap (25)	0.655 ± 0.079	0.556 ± 0.144	0.338 ± 0.151	–	0.169 ± 0.134
<i>M2 profile copy</i>					
All features (SHAP order)	0.410 ± 0.006	0.114 ± 0.001	0.036 ± 0.003	–	0.005 ± 0.001
Cheap only (SHAP order)	0.413 ± 0.003	0.107 ± 0.003	0.043 ± 0.002	–	0.011 ± 0.001
Strict cheap (SHAP order)	0.571 ± 0.003	0.260 ± 0.003	0.163 ± 0.004	0.140 ± 0.002	–
Structural only (SHAP order)	0.690 ± 0.000	0.606 ± 0.001	0.531 ± 0.005	–	0.496 ± 0.004
Random k of all 62	0.674 ± 0.075	0.572 ± 0.159	0.424 ± 0.169	–	0.248 ± 0.196
Random k of cheap (25)	0.655 ± 0.079	0.558 ± 0.141	0.343 ± 0.145	–	0.180 ± 0.130

Table 9: rev_models_10split

Model	Precision	Recall	F1	AP	AUROC	p (F1)
LightGBM	0.717 ± 0.022	0.710 ± 0.022	0.713 ± 0.006	0.761 ± 0.006	0.968 ± 0.002	–
XGBoost	0.718 ± 0.012	0.696 ± 0.019	0.707 ± 0.008	0.753 ± 0.008	0.964 ± 0.002	0.457
Random forest	0.722 ± 0.019	0.684 ± 0.021	0.702 ± 0.006	0.754 ± 0.007	0.969 ± 0.001	0.320
Logistic regression	0.146 ± 0.004	0.582 ± 0.039	0.234 ± 0.004	0.160 ± 0.009	0.832 ± 0.003	<0.001

Table 10: rev_operating_points

Threshold	Flagged	TP	FP	FP on list	Precision	Precision [†]
0.950	2,996	2,622	374	64	0.875	0.897
0.900	3,644	3,064	580	116	0.841	0.873
0.850	4,071	3,338	733	165	0.820	0.860
0.800	4,366	3,500	866	227	0.802	0.854
0.750	4,650	3,635	1,015	310	0.782	0.848
0.700	4,898	3,725	1,173	408	0.761	0.844
0.600 (val.)	5,346	3,885	1,461	556	0.727	0.831
0.500	5,807	4,040	1,767	707	0.696	0.817

Table 11: rev_per_category

Model	Category	Test Sybils	Precision	Recall	F1	AUROC
LightGBM	IxL	4,774.6	0.711 ± 0.020	0.771 ± 0.023	0.740 ± 0.004	0.975 ± 0.001
	IxI	469.2	0.886 ± 0.106	0.348 ± 0.078	0.492 ± 0.087	0.925 ± 0.010
	IxE	211.9	0.689 ± 0.106	0.148 ± 0.045	0.242 ± 0.065	0.910 ± 0.016
	No provider	7.3	0.870 ± 0.173	0.626 ± 0.148	0.716 ± 0.128	0.968 ± 0.028
XGBoost	IxL	4,774.6	0.713 ± 0.013	0.764 ± 0.019	0.738 ± 0.004	0.974 ± 0.001
	IxI	469.2	0.902 ± 0.053	0.254 ± 0.091	0.389 ± 0.113	0.914 ± 0.014
	IxE	211.9	0.708 ± 0.094	0.140 ± 0.039	0.232 ± 0.058	0.885 ± 0.016
	No provider	7.3	0.893 ± 0.142	0.626 ± 0.148	0.724 ± 0.117	0.969 ± 0.027

Table 12: rev_provision_ablation_by_category

Removed	All	IxL	IxI	IxE
Without all provision-network families	+0.008 $\pm$ 0.008	+0.003 $\pm$ 0.005	+0.114 $\pm$ 0.112	-0.038 $\pm$ 0.044
full model higher; p	7/10; 0.248	8/10; 0.507	9/10; 0.261	3/10; 0.338
Without Gas provider	+0.005 $\pm$ 0.006	+0.003 $\pm$ 0.004	+0.052 $\pm$ 0.091	+0.005 $\pm$ 0.019
full model higher; p	9/10; 0.341	8/10; 0.446	7/10; 0.516	6/10; 0.771
Without Gas provision tree	+0.001 $\pm$ 0.005	-0.001 $\pm$ 0.004	+0.036 $\pm$ 0.099	-0.008 $\pm$ 0.039
full model higher; p	7/10; 0.806	5/10; 0.769	6/10; 0.680	4/10; 0.808
Without Provision chain	+0.002 $\pm$ 0.002	-0.001 $\pm$ 0.002	+0.036 $\pm$ 0.067	+0.011 $\pm$ 0.014
full model higher; p	8/10; 0.391	4/10; 0.761	7/10; 0.544	7/10; 0.390

Table 13: rev_report_dependence

Model	Metric	Held out		Seen	Seen – held out	Seen higher	p
LightGBM	F1	0.369 ± 0.084	0.598 ± 0.024		$+0.229 \pm 0.076$	15/15	<0.001
	AP	0.305 ± 0.088	0.599 ± 0.033		$+0.294 \pm 0.077$	15/15	<0.001
	AUROC	0.891 ± 0.047	0.970 ± 0.006		$+0.079 \pm 0.042$	15/15	0.001
XGBoost	F1	0.363 ± 0.092	0.581 ± 0.033		$+0.218 \pm 0.081$	15/15	<0.001
	AP	0.305 ± 0.090	0.595 ± 0.036		$+0.290 \pm 0.075$	15/15	<0.001
	AUROC	0.888 ± 0.048	0.967 ± 0.009		$+0.080 \pm 0.043$	15/15	0.002

Table 14: rev_search_top

Rank	Configuration	Val. F1	Val. AP	Rounds
<i>XGBoost</i> (74 configurations)				
1	lr 0.05, depth 8, sub 0.7, col 1.0, mcw 1 *	0.7032	0.7552	1,679
2	lr 0.05, depth 10, sub 0.5, col 1.0, mcw 1	0.7025	0.7527	906
3	lr 0.05, depth 12, sub 0.5, col 0.8, mcw 1	0.7019	0.7543	547
4	lr 0.05, depth 8, sub 0.7, col 1.0, mcw 10	0.7016	0.7518	2,536
5	lr 0.1, depth 12, sub 0.7, col 0.8, mcw 1	0.7012	0.7525	297
6	lr 0.1, depth 10, sub 0.7, col 0.8, mcw 1	0.7012	0.7531	481
7	lr 0.05, depth 8, sub 0.7, col 1.0, mcw 5	0.7010	0.7541	1,927
8	lr 0.1, depth 8, sub 0.7, col 1.0, mcw 1	0.7010	0.7514	855
60	lr 0.3, depth 6, sub 1.0, col 1.0, mcw 1 (default)	0.6907	0.7422	671
<i>LightGBM</i> (48 configurations)				
1	leaves 127, lr 0.05, sub 0.7, λ 1.0 *	0.7067	0.7600	1,277
2	leaves 511, lr 0.05, sub 0.7, λ 0.0	0.7061	0.7543	270
3	leaves 255, lr 0.05, sub 0.7, λ 1.0	0.7047	0.7587	592
4	leaves 1023, lr 0.05, sub 0.7, λ 1.0	0.7039	0.7590	164
5	leaves 31, lr 0.05, sub 0.7, λ 1.0	0.7038	0.7558	4,749
6	leaves 31, lr 0.05, sub 0.7, λ 0.0	0.7036	0.7516	4,321
7	leaves 511, lr 0.1, sub 0.7, λ 1.0	0.7033	0.7535	156
8	leaves 511, lr 0.05, sub 0.7, λ 1.0	0.7025	0.7599	323
48	leaves 31, lr 0.1, sub 1.0, λ 0.0 (default)	0.6871	0.7376	2,059
<i>Random Forest</i> (24 configurations)				
1	mf 0.3, leaf 1, depth None *	0.7028	0.7515	—
2	mf 0.5, leaf 1, depth None	0.7021	0.7484	—
3	mf 0.7, leaf 1, depth None	0.7000	0.7423	—
4	mf sqrt, leaf 1, depth None	0.6990	0.7469	—
5	mf 0.5, leaf 5, depth None	0.6979	0.7424	—
6	mf 0.7, leaf 5, depth None	0.6976	0.7393	—
7	mf 0.3, leaf 5, depth None	0.6954	0.7426	—
8	mf 0.5, leaf 20, depth None	0.6815	0.7234	—

Table 15: rev_single_feature_sensitivity

Feature	Class	SHAP rank	Recall
l0_tx_time_span	time-bound	1	0.410 ± 0.006
n_l0_source_contracts	cheap	2	0.413 ± 0.003
earliest_l0_tx_time	time-bound	4	0.422 ± 0.003
l0_avg_stargate_swap	cheap	3	0.474 ± 0.002
earliest_tx_block_in	time-bound	10	0.517 ± 0.002
n_l0_source_chains	cheap	8	0.537 ± 0.003
latest_l0_tx_time	time-bound	5	0.545 ± 0.005
num_transactions_in	cheap	7	0.549 ± 0.003
max_tx_value_out	cheap	6	0.571 ± 0.003
time_span_in	time-bound	13	0.581 ± 0.003
gas_provision_block_number	time-bound	14	0.593 ± 0.001
tx_value_per_block_out	cheap	9	0.598 ± 0.003
min_tx_value_in	cheap	12	0.646 ± 0.004
n_l0_project_per_source_chain	cheap	15	0.646 ± 0.003
n_l0_projects	cheap	16	0.656 ± 0.003
No attack			0.711

Table 16: rev_taxonomy

Category	Addresses	Share	Sybils	Sybil rate
IxL	306,658	70.5%	15,921	5.19%
IxE	93,061	21.4%	698	0.75%
IxI	34,391	7.9%	1,566	4.55%
No provider	676	0.2%	26	3.85%
Total	434,786	100.0%	18,211	4.19%

Table 17: rev_temporal_holdout

Late	From	Sybil early	Sybil late	F1	Recall	AP	AUROC	F1 ^r	Recall ^r	AP ^r	AUROC ^r
20%	2023-12-17	5.16%	0.29%	0.008	0.004	0.060	0.729	0.483	0.328	0.446	0.935
30%	2023-08-20	5.55%	1.02%	0.169	0.094	0.276	0.887	0.738	0.695	0.774	0.978
40%	2023-06-27	5.11%	2.80%	0.213	0.142	0.231	0.894	0.727	0.765	0.807	0.983

Table 18: rev_union_split_appendix

Split	Partition	Rows	Sybils	In large comps	Reports
tree	Train	213,046	8,924	4,478	271
tree	Validation	91,305	3,824	1,968	229
tree	Test	130,435	5,463	2,768	239
union	Train	213,046	8,924	8,260	146
union	Validation	91,305	3,824	954	66
union	Test	130,435	5,463	0	99
<i>Union split, test</i>	<i>Model</i>	<i>F1</i>	<i>AP</i>	<i>AUROC</i>	
seed 42 (3 seeds)	XGBoost	0.266	0.220	0.818	–
seed 42 (3 seeds)	LightGBM	0.283	0.225	0.811	–
10 splits (1 seed)	LightGBM	0.257 ± 0.016	0.197 ± 0.017	0.771 ± 0.021	–
10 splits, tree split	LightGBM	0.713 ± 0.006	0.761 ± 0.006	0.968 ± 0.002	–